AI ASSISTANT CODING

ASSIGNMENT-16

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Batch:22

Task Descrption-1: (Design Database Schema for Student Management System)

Instructions:

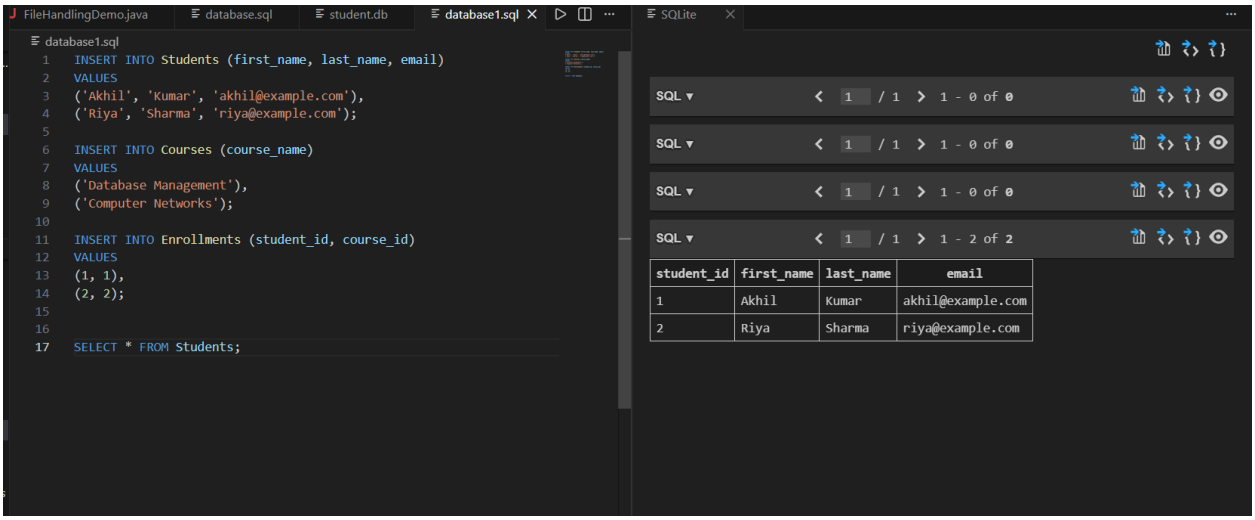
• Using AI tools, generate CREATE TABLE statements for a Student Management Database.

• Tables: Students, Courses, Enrollments.

• Include primary keys, foreign keys, and suitable data types.

Ensure the schema follows normalization rules (1NF, 2NF, 3NF).

Code:



Task Description- 2: (Insert Sample Records)

Instructions:

• Use AI tools to generate INSERT statements to populate the

tables.

• Insert at least 3 students, 4 courses, and 5 enrollment records.

Verify data using SELECT queries.

Code:

```sql
-
CREATE TABLE IF NOT EXISTS Students (
    student_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT,
    email TEXT UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS Courses (
    course_id INTEGER PRIMARY KEY AUTOINCREMENT,
    course_name TEXT NOT NULL
);

CREATE TABLE IF NOT EXISTS Enrollments (
    enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER,
    course_id INTEGER,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);



INSERT INTO Students (first_name, last_name, email)
VALUES
('Akhil', 'Kumar', 'akhil@example.com'),
('Riya', 'Sharma', 'riya@example.com'),
('Arjun', 'Reddy', 'arjun@example.com');

INSERT INTO Courses (course_name)
```

SQL ▾ 1 / 1 1 - 2 of 2

student_id	first_name	last_name	email
1	Akhil	Kumar	akhil@example.com
2	Riya	Sharma	riya@example.com

SQL ▾ 1 / 1 1 - 2 of 2

enrollment_id	student_id	course_id
1	1	1
2	2	2

SQL ▾ 1 / 1 1 - 2 of 2

course_id	course_name
1	Database Management
2	Computer Networks

Task Description- 3: (Perform CRUD

Operations) Instructions:

Use AI to generate SQL queries for basic CRUD operations:

- Retrieve all courses
- Update a student's email
- Delete an enrollment record
- Validate the changes using SELECT statements.

Code:

FileHandlingDemo.java | database.sql | student.db | database1.sql | SQLite

database1.sql

```sql
-
SELECT * FROM Courses;



UPDATE Students
SET email = 'akshaya_new@example.com'
WHERE student_id = 1;



SELECT * FROM Students WHERE student_id = 1;



DELETE FROM Enrollments
WHERE enrollment_id = 1;


-
SELECT * FROM Enrollments;


-
SELECT * FROM Students;
SELECT * FROM Courses;
SELECT * FROM Enrollments;
```

SQL ▾ 1 / 1 1 - 2 of 2

student_id	first_name	last_name	email
1	Akhil	Kumar	akhil@example.com
2	Riya	Sharma	riya@example.com

SQL ▾ 1 / 1 1 - 2 of 2

course_id	course_name
1	Database Management
2	Computer Networks

SQL ▾ 1 / 1 1 - 2 of 2

enrollment_id	student_id	course_id
1	1	1
2	2	2

```
SQL ▾                    1 / 1   1 - 0 of 0
UPDATE Students
SET email = 'john.newemail@example.com'
WHERE student_id = 1;
```

```
SQL ▾                    1 / 1   1 - 0 of 0
DELETE FROM Enrollments
WHERE enrollment_id = 1;
```

Task Description- 4: (Execute Aggregate Queries)

Instructions:

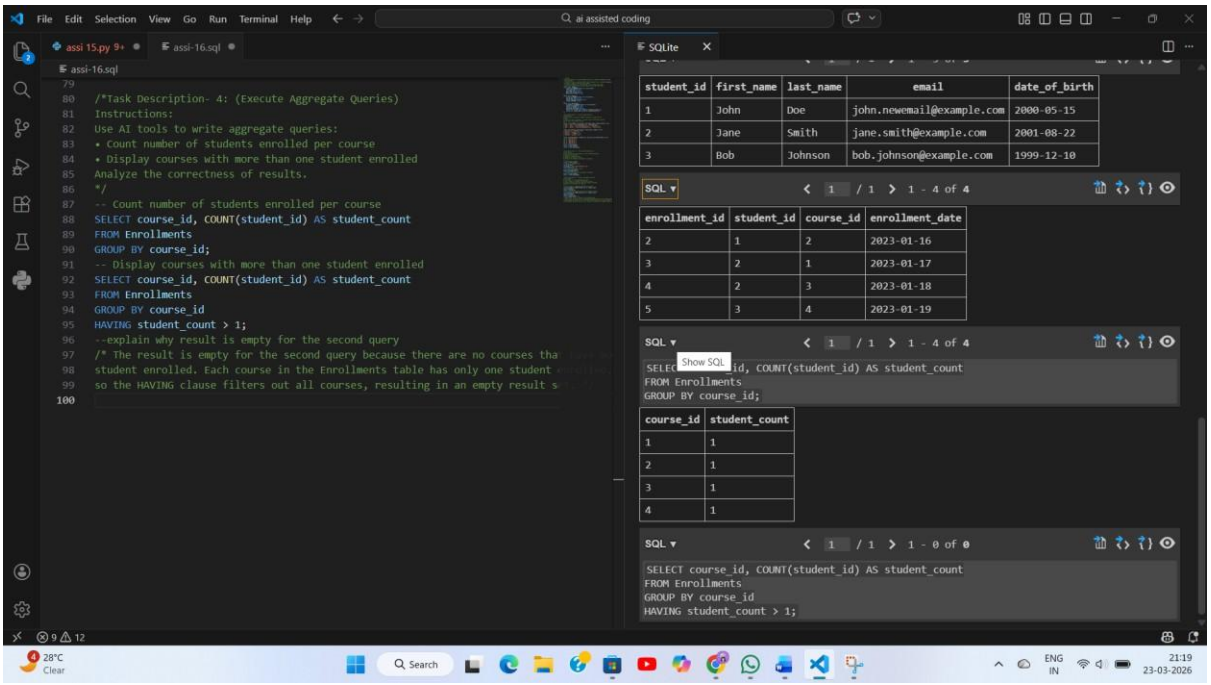
Use AI tools to write aggregate queries:

• Count number of students enrolled per course

• Display courses with more than one student enrolled

Analyze the correctness of results.

Code:



Task Description- 5: (Query Optimization)

Instructions:

• Use AI tools to analyze query performance.

• Rewrite inefficient sub queries using JOINs and test both queries

for correctness.

Code:

```sql
/*Task Description- 5: (Query Optimization)
Instructions:
• Use AI tools to analyze query performance.
• Rewrite inefficient sub queries using JOINs and test both queries
for correctness.*/
-- Inefficient subquery using IN
SELECT course_name
FROM Courses
WHERE course_id IN (SELECT course_id FROM Enrollments WHERE student_id = 2);
-- Optimized query using JOIN
SELECT c.course_name
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
WHERE e.student_id = 1;
```

3	2	1	2023-01-17
4	2	3	2023-01-18
5	3	4	2023-01-19

SQL ▾ 1 / 1 1 - 4 of 4

course_id	student_count
1	1
2	1
3	1
4	1

SQL ▾ 1 / 1 1 - 0 of 0

SQL ▾ 1 / 1 1 - 2 of 2

```sql
SELECT course_name
FROM Courses
WHERE course_id IN (SELECT course_id FROM Enrollments WHERE student_id = 2);
```

course_name
Mathematics
Chemistry

SQL ▾ 1 / 1 1 - 1 of 1

```sql
SELECT c.course_name
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
WHERE e.student_id = 1;
```

course_name
Physics